

UNIT-V

Vector Integration

~~Line Integral~~ If $\vec{F}$ & $\vec{u}$ are 2 vector point function such that $\frac{\partial \vec{F}(t)}{\partial t} = \vec{u}(t)$ Then $\vec{F}$ is called integral of $\vec{u}(t)$ denoted by $\vec{F}(t) = \int \vec{u}(t) dt$

Line Integral: The line integral of a vector field $\vec{F}$ is denoted by $\int_a^b \vec{F} \cdot d\vec{R}$

i.e. line integral means integral along a curve

→ The line integral gives the work done by the force $\vec{F}$ in moving a particle from one point a to another point b along a path curve c

→ If the curve is closed then the line integral is

$$\oint_C \vec{F} \cdot d\vec{R}$$

→ In fluid-dynamics If $\vec{F}$ represents the velocity of flow of fluid then integral along c , $\oint_C \vec{F} \cdot d\vec{R}$ gives the circulation of $\vec{F}$ along c

$$\vec{S} = (5)\hat{i} + (2)\frac{\vec{S}}{\sqrt{5}} + (25)\frac{\vec{S}}{\sqrt{5}}$$

$$(15)\frac{\vec{S}}{\sqrt{5}} + (2)\frac{\vec{S}}{\sqrt{5}} + (10)\frac{\vec{S}}{\sqrt{5}} = 17\vec{R}$$

$$\vec{S} = (10\sqrt{5})\hat{i} + (2\sqrt{5})\hat{j} + (10\sqrt{5})\hat{k}$$

$$(2)(10)\hat{j} = 20\hat{j}$$

Problems on Line Integral

(Q1) If $\vec{F} = (5xy - 6x^2)\vec{i} + (2y - 4x)\vec{j}$, evaluate $\int_C \vec{F} \cdot d\vec{R}$ along the curve C in xy -plane, $y = x^3$ from $(1,1)$ to $(2,8)$

Sol Given, $\vec{F} = (5xy - 6x^2)\vec{i} + (2y - 4x)\vec{j}$

$$\text{Here } \vec{R} = x\vec{i} + y\vec{j}$$

$$d\vec{R} = dx\vec{i} + dy\vec{j}$$

$$\text{Given } C: y = x^3 \Rightarrow dy = 3x^2 dx$$

$$\& x: 1 \text{ to } 2$$

$$\begin{aligned} \therefore \int_C \vec{F} \cdot d\vec{R} &= \int_C (5xy - 6x^2) dx + (2y - 4x) dy \\ &= \int_{x=1}^2 (5x^4 - 6x^2) dx + (2x^3 - 4x) 3x^2 dx \end{aligned}$$

$$= \left[\frac{5x^5}{5} - \frac{6x^3}{3} + \frac{6x^6}{6} - \frac{12x^4}{4} \right]_1^2$$

$$= (32-1) - 2(8-1) + (64-1) - 3(16-1)$$

$$= 31 - 14 + 63 - 45 = 94 - 59 = 35$$

Green's Theorem in a plane

Let $\phi(x, y)$ & $\psi(x, y)$ two continuous partial derivative ϕ_x & ψ_x in a region of (x, y) plane bounded by a closed curve 'C' then

$$\oint_C \phi dx + \psi dy = \iint_R \left(\frac{\partial \psi}{\partial x} - \frac{\partial \phi}{\partial y} \right) dx dy$$

NOTE: This theorem is a relation b/w line integral to a double integral 'R' enclosed by the curve 'C'

(Pb) Evaluate $\oint_C (x^2 - 2xy) dx + (x^2 y + 3) dy$ around the boundary of the region defined by $y = \sqrt{8x}$ & $x = 2$ using green's theorem

Sol Given that, $\oint_C (x^2 - 2xy) dx + (x^2 y + 3) dy$

$\phi = x^2 - 2xy$	$\psi = x^2 y + 3$
$\text{now } \frac{\partial \phi}{\partial x} = 2x - 2y$	$\frac{\partial \psi}{\partial y} = x^2$
$\text{now } \frac{\partial \phi}{\partial y} = -2x$	$\frac{\partial \psi}{\partial x} = 2xy$

By Green's theorem; $\oint_C (x^2 - 2xy) dx + (x^2 y + 3) dy$

$$= \iint_R (+2x - 2xy) dy dx$$

$$= \iint_R (2xy + 2x) dx dy$$

Here R: $0 \leq x \leq 2$

$$-\sqrt{8x} \leq y \leq \sqrt{8x}$$

$$\begin{aligned} \therefore \oint_C (x^2 - 2xy) dx + (x^2 y + 3) dy \\ = \int_0^2 \int_{-\sqrt{8x}}^{\sqrt{8x}} (2xy + 2x) dy dx \end{aligned}$$

$$= \int_0^2 (2x)(y) \frac{dy}{dx} dx$$

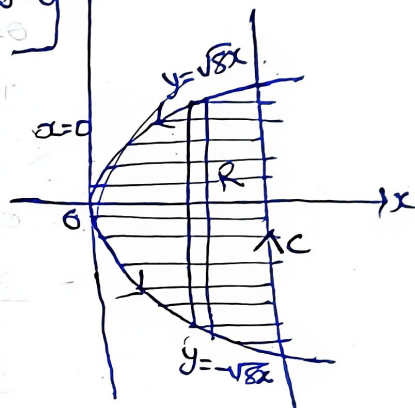
$$\left[\because \int_{-\sqrt{8x}}^{\sqrt{8x}} y dy = 0 \right]$$

$$= 2 \int_0^2 x (2\sqrt{8x}) dx$$

$$= 4\sqrt{8} \int_0^2 x^{3/2} dx$$

$$= \frac{8\sqrt{8}}{5} \left(\frac{5}{2} \right) = \frac{8(2\sqrt{2})}{5} \cdot 2 \cdot \sqrt{2}$$

$$= \frac{128}{5}$$



(P6) Applying Green's theorem to evaluate the integral $\oint_C (2x^2 - y^2) dx + (x^2 + y^2) dy$ where 'C' is the boundary of the area enclosed by the curves x-axis & the upper half of the circle $x^2 + y^2 = a^2$

sol Given that: $\oint_C (2x^2 - y^2) dx + (x^2 + y^2) dy$

$$\text{now } \phi = 2x^2 - y^2$$

$$\psi = x^2 + y^2$$

$$\frac{\partial \phi}{\partial y} = -2y$$

$$\frac{\partial \psi}{\partial x} = 2x$$

$$\oint_C (2x^2 - y^2) dx + (x^2 + y^2) dy = \iint_R (2x + 2y) dx dy$$

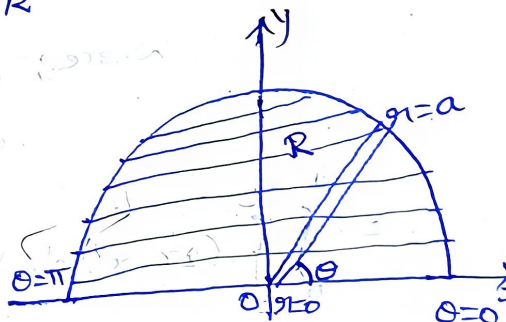
$$R: 0 \leq r \leq a$$

$$0 \leq \theta \leq \pi$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx dy = r dr d\theta$$



$$\begin{aligned}
 &= 2 \int_{\theta=0}^{\pi} \int_{x=0}^a (x \cos \theta + y \sin \theta) \\
 &= 2 \int_{\theta=0}^{\pi} (\cos \theta + \sin \theta) \left(\frac{x^3}{3} \right)_0^a d\theta \\
 &= \frac{2a^3}{3} [\sin \theta - \cos \theta]_0^{\pi} \\
 &= \frac{4a^3}{3}
 \end{aligned}$$

⑩ using Green's theorem evaluate integral along closed curve 'c'. $\oint_C (x^2y)dx + x^2y dy$ where 'c' is the boundary of the triangle $(0,0), (1,0), (1,1)$

Sol Given that

$$\oint_C (x^2y)dx + x^2y dy$$

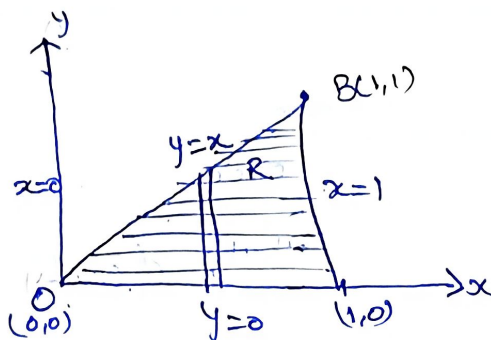
$$= \oint$$

$$\phi = x^2y$$

$$\frac{\partial \phi}{\partial y} = x^2$$

$$\psi = x^2$$

$$\frac{\partial \psi}{\partial x} = 2x$$



$$\oint_C (x^2y)dx + x^2y dy = \iint_R (2x - x^2) dx dy$$

where; $0 \leq x \leq 1$

$0 \leq y \leq x$

$$= \int_0^1 \int_0^x (2x - x^2) dx dy$$

$$= 2 \int_0^1 (2x - x^2) [y]_0^x dx = \int_0^1 (2x^2 - x^3) dx$$

$$= \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{2}{3} - \frac{1}{4} = \frac{8-3}{12} = \frac{5}{12}$$

(P6) Verify Green's theorem for the integral $\oint_C x^2 y dx + x^2 dy$
 $\oint_C (2y + y^2) dx + x^2 dy$ where 'C' is bounded by $y=x$ & $y=x^2$

sol Given that

$$\oint_C (2y + y^2) dx + x^2 dy$$

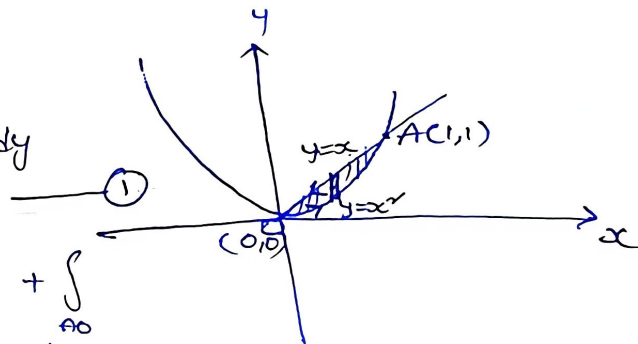
$$\begin{array}{l|l} \phi = 2y + y^2 & \psi = x^2 \\ \frac{\partial \phi}{\partial y} = 2 + 2y & \frac{\partial \psi}{\partial x} = 2x \end{array}$$

Given that $y=x$ & $y=x^2$

Points of intersection are $(0,0)$, $(1,1)$

By Green's theorem,

$$\oint_C (2y + y^2) dx + x^2 dy = \iint_R (2 - 2y) dx dy$$



$$\text{LHS of (1)} \quad \oint_C (2y + y^2) dx + x^2 dy = \int_{OA} + \int_{AO}$$

(1) Along Arc OA

$$y=x^2$$

$$\Rightarrow dy = 2x dx ; x: 0 \text{ to } 1$$

$$\therefore \int_{OA} (2y + y^2) dx + x^2 dy = \int_{x=0}^1 (2x^2 + x^4) dx + x^2 (2x dx)$$

$$= \int_{x=0}^1 (3x^3 + x^4) dx = \left(\frac{3x^4}{4} + \frac{x^5}{5} \right)_0^1$$

$$= \frac{19}{20}$$

② Along the line AOB: $y=x$

$$\Rightarrow dy = dx$$

& $x: 1 \text{ to } 0$

$$\int_{AO} (xy + y^2) dx + x^2 dy = \int_{x=1}^0 3x^2 dx$$

(line)

$$= 3 \left(\frac{x^3}{3} \right)_1^0 = -1$$

$$\therefore \text{L.H.S of } \textcircled{1} = \frac{19}{20} + (-1)$$

$$= \frac{-1}{20}$$

$$\text{R.H.S of } \textcircled{1} = \iint_R (x-2y) dx dy$$

$$= \int_{x=0}^1 \int_{y=x}^x (x-2y) dx dy$$

$$= \int_0^1 \left[x(y) \right]_{x^2}^x - \left[\frac{2y^2}{2} \right]_{x^2}^x dy$$

$$= \int_0^1 (x(x-x^2) - x^2 + x^2) dx$$

$$= \int_0^1 (-x^3 + x^4) dx$$

$$\left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1$$

$$= \frac{1}{5} - \frac{1}{4} - 0 + 0$$

$$= \frac{-1}{20}$$

Q6) $\oint_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$ at $x=0, y=0$ & $x+y=1$

sol Given that,

$$\oint_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$$

Here $\phi = 3x^2 - 8y^2$ $\psi = 4y - 6xy$

$$\frac{\partial \phi}{\partial y} = -16y \quad \frac{\partial \psi}{\partial x} = -6y$$

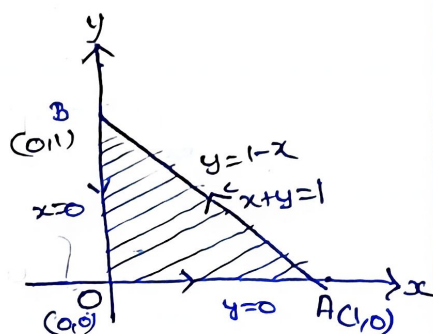
By Green's theorem,

$$\oint_C (3x^2 - 8y^2) dx + (4y - 6xy) dy = \iint_R 10y \, dx \, dy \quad \text{--- (1)}$$

Given curves $x=0, y=0$ & $x+y=1$

Now, LHS of (1) = $\oint_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$

$$= \int_{OA} + \int_{AB} + \int_{BO}$$



① Along OA: $y=0 \Rightarrow dy=0$
& $x=0$ to 1

$$\therefore \int_{OA} (3x^2 - 8y^2) dx + (4y - 6xy) dy = \int_{x=0}^1 3x^2 dx = \left[\frac{3x^3}{3} \right]_0^1 = 1$$

② Along AB: $x+y=1$

$$\Rightarrow y=1-x$$

$$dy = -dx$$

& $x: 1$ to 0

$$\therefore \int_{AB} (3x^2 - 8y^2) dx + (4y - 6xy) dy$$

$$= \int_{x=1}^0 3x^2 - 8(1-x)^2 dx + (4(1-x) - 6x(1-x)) dy$$

$$= \int_{x=1}^0 (3x^2 - 8 - 8x^2 + 16x + 4 - 4x + 6x - 6x^2) dx$$

$$= \int_{x=1}^0 (-11x^2 + 26x + 9) dx$$

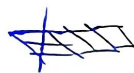
$$= \left[-\frac{11}{3}x^3 + \frac{26}{2}x^2 + 9x \right]_1^0$$

$$= \frac{11}{3} - 13 + 9 = \frac{11}{3} - 4$$

$$= \frac{11}{3} - \frac{12}{3} = \frac{11-12}{3} = -\frac{1}{3}$$

iii) Along B0: $x=0 \Rightarrow dx=0$

& $y: 1 \text{ to } 0$



$$\therefore \int_{B0} (3x^2 - 8xy) dx + (4y - 2xy) dy$$

$$= \int_{y=1}^0 4y dy = \left(4 \frac{y^2}{2} \right)_1^0 = -2$$

$$\therefore \text{L.H.S of } \textcircled{1} = 1 + \frac{8}{3} - 2 = \frac{5}{3}$$

R.H.S of $\textcircled{1} = \iint_R 10y dy dx$

$$R: 0 \leq x \leq 1$$

$$0 \leq y \leq 1-x$$

$$= 10 \int_0^1 \int_0^{1-x} y dy dx$$

$$= 10 \int_0^1 \left(\frac{y^2}{2} \right)_0^{1-x} dx$$

$$= \frac{10}{2} \int_0^1 (1-x)^2 dx$$

$$= \frac{5}{3}$$

Qb) verify Green's theorem for the integral

$\oint_C (y - \sin x) dx + \cos x dy$ where C is the boundary of the region

formed by the lines $y=0$, $x=\frac{\pi}{2}$ & $y=\frac{2x}{\pi}$

sol Given that,

$$\oint_C (y - \sin x) dx + \cos x dy$$

Now, $\phi = y - \sin x$

$\psi = \cos x$

$$\frac{\partial \phi}{\partial y} = 1$$

$$\frac{\partial \psi}{\partial x} = -\sin x$$

By Green's theorem,

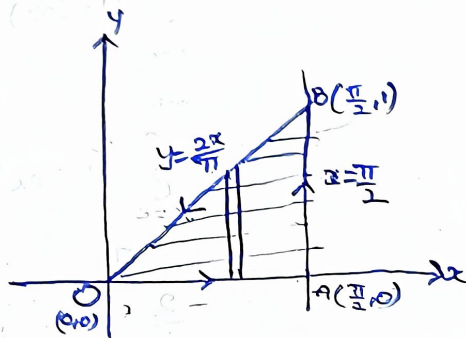
$$\oint_C (y - \sin x) dx + \cos x dy = \iint_R -(\sin x + 1) dx dy \quad \text{--- ①}$$

Given lines are $y=0$, $x=\frac{\pi}{2}$ & $y=\frac{2x}{\pi}$

Now LHS of ① = $\int_{OA} + \int_{AB} + \int_{BO}$

① Along OA: $y=0 \Rightarrow dy=0$
& $x=0$ to $\frac{\pi}{2}$

$$\begin{aligned} & \therefore \int (y - \sin x) dx + \cos x dy \\ &= \int_{x=0}^{\pi/2} -\sin x dx = (\cos x)_0^{\pi/2} = -1 \end{aligned}$$



② Along AB: $x=\frac{\pi}{2} \Rightarrow dx=0$
& $y: 0$ to 1

$$\int_{AB} (y - \sin x) dx + \cos x dy = \int_0^1 y \sin \cos \frac{\pi}{2} dy = 0$$

③ Along BO: $y=\frac{2x}{\pi} \Rightarrow dy = \frac{2}{\pi} dx$

$$\int_{\pi/2}^0 \left(\frac{2x}{\pi} - \sin x \right) dx + \cos x \left(\frac{2}{\pi} dx \right)$$

$$= \left[\frac{2}{\pi} \cdot \frac{x^2}{2} + \cos x + \frac{2}{\pi} \sin x \right]_{\pi/2}^0$$

$$= 1 - \frac{\pi}{4} + \frac{2}{\pi}$$

∴

$$\text{L.H.S of } \textcircled{1} = -1 + 1 - \frac{\pi}{4} - \frac{2}{\pi}$$

$$= -\frac{\pi}{4} - \frac{2}{\pi}$$

$$\text{R.H.S of } \textcircled{1} = - \iint_R (1 + \sin x) dx dy$$

$$= - \int_{x=0}^{\pi/2} \int_0^{\frac{2x}{\pi}} (1 + \sin x) dy dx$$

$$= - \int_{x=0}^{\pi/2} (1 + \sin x) (y)_0^{\frac{2x}{\pi}} dx$$

$$= - \int_{x=0}^{\pi/2} (1 + \sin x) \left(\frac{2x}{\pi} \right) dx$$

$$= - \frac{2}{\pi} \int_{x=0}^{\pi/2} (x + x \sin x) dx$$

$$= - \frac{2}{\pi} \left[\left[\frac{x^2}{2} \right]_0^{\pi/2} + \left[x(-\cos x) + \sin x \right]_0^{\pi/2} \right]$$

$$= -\frac{\pi}{4} - \frac{2}{\pi}$$

$$\text{L.H.S of } \textcircled{1} = \text{R.H.S of } \textcircled{1}$$

∴ Hence Green's theorem is verified

Surface Integral

Any integral which is to be evaluated is called a surface integral.

Let $\vec{F}$ be a vector function over the surface 'S' & $\hat{n}$ be the unit normal to the surface then $\int_S \vec{F} \cdot \hat{n} ds$ is the surface integral.

$$= \iint_{R_{xy}} \vec{F} \cdot \hat{n} \frac{dx dy}{|\hat{n} \cdot \hat{k}|}$$

Evaluate the surface integral

A surface integral can be evaluated by projecting to given surface into any one of the co-ordinate planes

i) If the projection of S is on xy-plane

then
$$\int_S \vec{F} \cdot \hat{n} ds = \iint_{R_{xy}} \vec{F} \cdot \hat{n} \frac{dx dy}{|\hat{n} \cdot \hat{k}|}$$

ii) If the projection of S is on xz-plane

then
$$\int_S \vec{F} \cdot \hat{n} ds = \iint_{R_{xz}} \vec{F} \cdot \hat{n} \frac{dx dz}{|\hat{n} \cdot \hat{j}|}$$

iii) If the projection of 'S' is on yz-plane

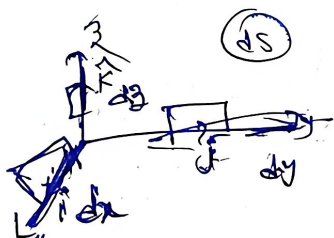
then
$$\int_S \vec{F} \cdot \hat{n} ds = \iint_{R_{yz}} \vec{F} \cdot \hat{n} \frac{dy dz}{|\hat{n} \cdot \hat{i}|}$$

NOTE:

* $\int_S \vec{F} \cdot \hat{n} ds$ represents the flux of the physical quantity moving over the surface S & $\hat{n}$ be the unit normal to the surface

* $\oint_S \vec{F} \cdot \hat{n} ds$ is the surface integral over the closed surface

'S'



Q6 Evaluate $\int_S \vec{F} \cdot \hat{n} \, ds$, where $\vec{F} = (x+y)\vec{i} - 2x\vec{j} + 2yz\vec{k}$
 S is the surface of the plane $2x+y+2z=6$ in the 1st octant

Sol: Given $\vec{F} = (x+y)\vec{i} - 2x\vec{j} + 2yz\vec{k}$

Given surface is ' S ' $= 2x+y+2z=6$

Now, $\nabla S = \text{Normal to 'S'}$

$$= 2\vec{i} + \vec{j} + 2\vec{k}$$

Let $\hat{n}$ be the unit normal to S then

$$\hat{n} = \frac{\nabla S}{|\nabla S|} = \frac{2\vec{i} + \vec{j} + 2\vec{k}}{\sqrt{4+1+4}} = \frac{2\vec{i} + \vec{j} + 2\vec{k}}{3}$$

Now $\vec{F} \cdot \hat{n} = \frac{2}{3}(x+y) + (-\frac{2}{3})x + \frac{4yz}{3}$

$$= \frac{2y}{3} + \frac{4y}{3} \left[\frac{6-2x-y}{2} \right]$$

$$= \frac{2y}{3} + \frac{4y}{3} \left[3 - x - \frac{y}{2} \right]$$

$$= \frac{2y}{3} + 4y - \frac{4xy}{3} - \frac{2y^2}{3}$$

On xy -plane,

$$\int_S \vec{F} \cdot \hat{n} \, ds = \iint_R \vec{F} \cdot \hat{n} \frac{dx \, dy}{|\hat{n} \cdot \vec{k}|}$$

$$= \int_{x=0}^3 \int_0^{6-2x} \frac{2}{3}(6y-2xy) \frac{dx \, dy}{\frac{2}{3}}$$

$$= \int_{x=0}^3 (6-2x) \left[\frac{y^2}{2} \right]_0^{6-2x} dy$$

$$= \frac{1}{2} \int_{x=0}^3 (6-2x)(6-2x)^2 dx$$

$$\frac{1}{2} \int_0^1 \int_0^1 \sqrt{1+4x^2+4y^2} \, dx \, dy$$

$$= \frac{1}{2} \int_0^1 \left[\frac{(1+4x^2)^{3/2}}{6} \right]_0^1 \, dy$$

$$= \frac{1}{16} \int_0^1 (1+4x^2)^{3/2} \, dy = \frac{1}{16} \left(\frac{1+4x^2}{2} \right)^{3/2} \Big|_0^1 = 81$$

⑧ If $\vec{F} = xy\vec{i} - x^2\vec{j} + x^2\vec{k}$ & S is the surface of parabolic cylinder $z = x^2$ in the first octant $y = 0$ to $y = 1$ & $x = 0$ to $x = 1$ evaluate $\int_S \vec{F} \cdot \vec{n} \, ds$

on

$$\frac{2x^2}{4} = \frac{2x^2 + 4x^2}{4}$$

$$\frac{4x^2}{4}$$

$$\frac{4x^2}{4}$$

$$1 - \frac{2x^2}{4}$$

to

Stokes's theorem:

Let $\vec{F}$ be

continuous

$\rightarrow$ A $\vec{F}$ be $\times$ define and over S bounded by a closed curve C

$$\oint_C \vec{F} \cdot d\vec{R} = \int_S \text{curl } \vec{F} \cdot \hat{n} \cdot ds \quad ; \text{ where } \hat{n} \text{ is the unit normal drawn outward to the surface}$$

Note: It is a relation b/w a closed curve integral to a open surface integral

(Pb) verify the Stokes's theorem of the vector field

$$\vec{F} = (2x-y)\vec{i} - yz\vec{j} - y^2z\vec{k}$$

over the bounded by its projection on xy plane

$$x^2 + y^2 + z^2 = 1$$

Sol Given that,

$$\vec{F} = (2x-y)\vec{i} - yz\vec{j} - y^2z\vec{k}$$

$$\text{Here } \vec{F} \cdot d\vec{R} = (2x-y)dx - yzdy - y^2zdz$$

By Stokes's theorem;

$$\text{L.H.S} = \oint_C \vec{F} \cdot d\vec{R} = \int_S \text{curl } \vec{F} \cdot \hat{n} \cdot ds$$

$$\text{L.H.S} = \oint_C (2x-y)dx - yzdy - y^2zdz$$

$$= \oint_C (2x-y)dx \quad \left(\because \text{on } xy\text{-plane} \right)$$

$$\text{Put } x = \cos\theta \quad ; \quad y = \sin\theta$$

$$dx = -\sin\theta d\theta$$

$$\oint \theta: 0 \text{ to } 2\pi$$

$$\text{then } \oint_C (2x-y)dx$$

$$= \int_{\theta=0}^{2\pi} (2\cos\theta - \sin\theta) d\theta$$

$$= \int_{\theta=0}^{2\pi} (2\cos\theta - \sin\theta) d\theta$$

$$= \left[2\sin\theta + \cos\theta \right]_0^{2\pi}$$

$$= (2\sin 2\pi + \cos 2\pi) - (2\sin 0 + \cos 0)$$

$$= (2 \cdot 0 + 1) - (0 + 1)$$

$$= 0$$

$$\text{Now } \text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz & -y^2z \end{vmatrix}$$

$$= \vec{i} \left[\frac{\partial}{\partial y}(-y^2z) - \frac{\partial}{\partial z}(-yz) \right] - \vec{j} \left[\frac{\partial}{\partial x}(-y^2z) - \frac{\partial}{\partial z}(2x-y) \right] + \vec{k} \left[\frac{\partial}{\partial x}(-yz) - \frac{\partial}{\partial y}(2x-y) \right]$$

$$= \vec{i} [-2yz - (-y)] - \vec{j} [0 - 0] + \vec{k} [-y - (-1)]$$

$$= \vec{i} (-2yz + y) + \vec{k} (-y + 1)$$

then $\oint_C (2x-y) dx$

$$= \int_{\theta=0}^{2\pi} (2\cos\theta - \sin\theta) - \sin\theta d\theta$$

$$= \int_{\theta=0}^{2\pi} (-\sin 2\theta) d\theta + \int_{\theta=0}^{2\pi} \left(\frac{1-\cos 2\theta}{2} \right) d\theta$$

$$= \left[\frac{\cos 2\theta}{2} \right]_0^{2\pi} + \int_{\theta=0}^{2\pi} \frac{1}{2} d\theta - \int_{\theta=0}^{2\pi} \frac{\cos 2\theta}{2} d\theta$$

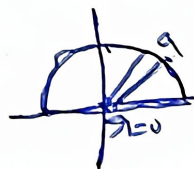
$$= \left(\frac{\cos 2\theta}{2} \right)_0^{2\pi} + \frac{1}{2}(\theta)_0^{2\pi} + \left(\frac{\sin 2\theta}{4} \right)_0^{2\pi}$$

$$= \frac{\cos 4\pi}{2} - \frac{1}{2} + \frac{1}{2}(2\pi) + \frac{\cos 4\pi}{4} - 0$$

$$= 0 + \frac{1}{2}(2\pi) + 0$$

$$= \pi$$

Now $\text{Curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (2x-y) & -y\sqrt{z} & -y\sqrt{z} \end{vmatrix}$



$$= \vec{i} \left[\frac{\partial}{\partial x} (-y\sqrt{z}) \right]$$

$$= \vec{i} \left[-\frac{\partial}{\partial y} (y\sqrt{z}) + \frac{\partial}{\partial z} (y\sqrt{z}) \right] + \vec{j} \left[\frac{\partial}{\partial x} (y\sqrt{z}) - \frac{\partial}{\partial z} (2x-y) \right]$$

$$+ \vec{k} \left[-\frac{\partial}{\partial x} (y\sqrt{z}) - \frac{\partial}{\partial y} (2x-y) \right]$$

$$= 2x\vec{i} + 2y\vec{j} + 2\sqrt{z}\vec{k}$$

Here $\hat{n}$ is unit vector normal to S'

$$= \frac{\nabla S}{|\nabla S|} = \frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = x\hat{i} + y\hat{j} + z\hat{k}$$

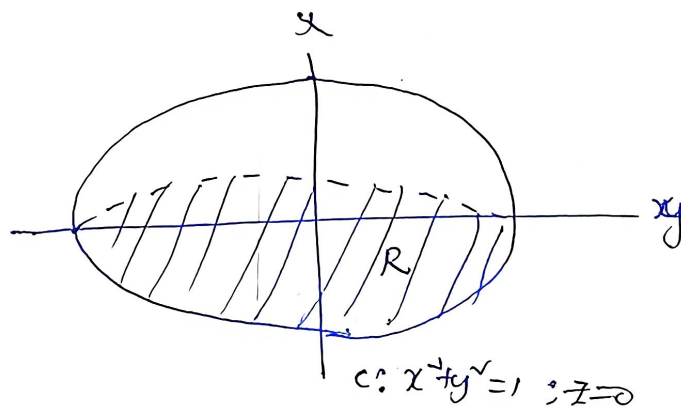
$$R.H.S \neq (1) = \int_S \text{curl } \vec{F} \cdot \hat{n} \, dS$$

$$= \iint_R z \cdot \frac{dx dy}{|\hat{n} \cdot \hat{k}|} = \iint_R \cancel{z} \cdot \frac{dx dy}{\cancel{z}} = \iint_R dx dy$$

$$\Rightarrow \iint_R dx dy = \text{Area of } C: x^2 + y^2 = 1$$

$$= \pi r^2 = \pi(1)^2 = \pi$$

$$L.H.S = R.H.S$$



(P6) Verify Stokes theorem $\vec{F} = (y - z + 2)\hat{i} + (yz + 4)\hat{j} - xz\hat{k}$

Where S is the surface of the cube $x=0; y=0; z=0;$

$x=2; y=2; z=2$. above the xy plane.

Sol Given that,

$$\vec{F} = (y - z + 2)\hat{i} + (yz + 4)\hat{j} - xz\hat{k}$$

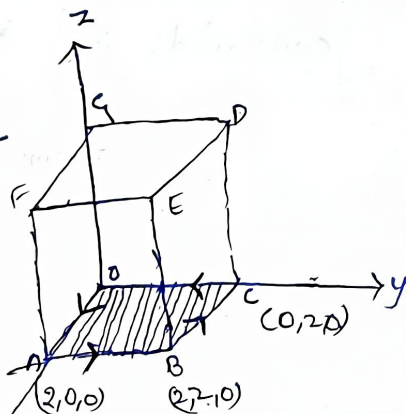
— here the open surface

consisting of 5 faces of the cube

$$x=0; y=0; z=0,$$

$$x=2; y=2; z=2.$$

The closed curve bounded by a open surface is OABCO



$$\therefore \oint_C \vec{F} \cdot d\vec{r} = \int_{OA} + \int_{AB} + \int_{BC} + \int_{CO}$$

(i) Along OA :

$$y=0, z=0$$

$$\& x: 0 \text{ to } 2$$

$$\therefore \int_{OA} \vec{F} \cdot d\vec{r} = \int_0^2 2 dx = 4$$

(ii) Along AB :

$$x=2, dx=0$$

$$z=0, dz=0$$

$$\therefore \int_{AB} \vec{F} \cdot d\vec{r} = \int_0^2 4 dy = 8$$

(iii) Along BC : $y=2; dy=0$

$$z=0, dz=0$$

$$x: 2 \text{ to } 0$$

$$\int_{BC} \vec{F} \cdot d\vec{r} = \int_2^0 4 dx = -8$$

(iv) Along CO :

$$x=0, dx=0$$

$$z=0, dz=0$$

$$y: 2 \text{ to } 0$$

$$\therefore \int_{CO} \vec{F} \cdot d\vec{r} = \int_2^0 4 dy = -8$$

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = 4 + 8 - 8 - 8 = -4$$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_S \text{curl } \vec{F} \cdot \hat{n} dS \quad \text{--- (1)}$$

$$\text{curl } \vec{F} = -y\vec{i} + (z-1)\vec{j} - \vec{k}$$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (1-3x) & (1-2+y) & -xz \end{vmatrix}$$

$$= \vec{i}[0] - \vec{j}[-1+1] + \vec{k}[1]$$

$$= \vec{k}$$

$$\boxed{\text{curl } \vec{F} = \vec{k}}$$

$$\iint_S \text{curl} \vec{F} \cdot \hat{n} \, dS = \iint_S \vec{R} \cdot \vec{R} \frac{\partial x \, dy}{|\vec{R} \cdot \vec{R}|}$$

$$= \pi R^2 = \pi(1) = \pi$$

here R.H.S = $\int_S \text{curl } \vec{F} \cdot \hat{n} \, ds = \int_{S_1} + \int_{S_2} + \int_{S_3} + \int_{S_4} + \int_{S_5}$
 (ABEF) (BCDE) (CDEF) (CDAE) (DAFC)

(i) For the Face S_1 : $\hat{n} = \hat{i}$; $ds = \frac{dydz}{|\hat{n} \cdot \hat{i}|} = dydz$

$$\therefore \int_{S_1} \text{curl } \vec{F} \cdot \hat{n} \, ds = \int_0^2 \int_0^2 (-y) \, dy \, dz = -2 \left(\frac{y^2}{2} \right)_0^2 = -4$$

(ii) For the Face S_2 : $\hat{n} = \hat{j}$; $ds = \frac{dx dz}{|\hat{n} \cdot \hat{j}|} = dx dz$

$$\therefore \int_{S_2} \text{curl } \vec{F} \cdot \hat{n} \, ds = \int_0^2 \int_0^2 (z-1) \, dx \, dz$$

$$= 2 \left(\frac{z}{2} - z \right)_0^2 = 2(2-2) = 0$$

(iii) For the Face S_3 : $\hat{n} = \hat{k}$; $ds = \frac{dx dy}{|\hat{n} \cdot \hat{k}|} = dx dy$

$$\therefore \int_{S_3} \text{curl } \vec{F} \cdot \hat{n} \, ds = \int_0^2 \int_0^2 (-1) \, dx \, dy = -4$$

(iv) For the Face S_4 : $\hat{n} = -\hat{i}$; $ds = dy dz$

$$\therefore \int_{S_4} \text{curl } \vec{F} \cdot \hat{n} \, ds = \int_0^2 \int_0^2 y \, dy \, dz = 2 \left(\frac{y^2}{2} \right)_0^2 = 4$$

(v) For the Face S_5 : $\hat{n} = -\hat{j}$; $ds = dx dz$

$$\therefore \int_{S_5} \text{curl } \vec{F} \cdot \hat{n} \, ds = \int_0^2 \int_0^2 (1-z) \, dx \, dz$$

$$= 2 \left[z - \frac{z^2}{2} \right]_0^2 = 2[2-2] = 0$$

$$\therefore \int_S \text{curl } \vec{F} \cdot \hat{n} \, ds = -4 + 0 - 4 + 4 + 0 = -4$$

$\therefore \text{L.H.S} = \text{R.H.S}$ Hence Stokes theorem is verified

(b) Evaluate by Stokes - theorem

$$\oint_C (x+y)dx + (2x-z)dy + (y+z)dz$$

where C is the boundary of the Δ^c with vertices $(0,0,0)$ & $(1,0,0)$ & $(1,1,0)$

so Given that,

$$\oint_C (x+y)dx + (2x-z)dy + (y+z)dz$$

$$\Rightarrow \vec{F} = (x+y)\vec{i} + (2x-z)\vec{j} + (y+z)\vec{k}$$

$$\text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x+y) & (2x-z) & (y+z) \end{vmatrix}$$

$$= \vec{i} \left[\frac{\partial}{\partial y}(y+z) - \frac{\partial}{\partial z}(2x-z) \right] - \vec{j} \left[\frac{\partial}{\partial x}(y+z) - \frac{\partial}{\partial z}(x+z) \right]$$

$$+ \vec{k} \left[\frac{\partial}{\partial x}(2x-z) - \frac{\partial}{\partial y}(x+y) \right]$$

$$= 2\vec{i} + \vec{k}$$

Here S is the bounded by $C: OABO$ Δ^c

so that $\vec{n} = \vec{k}$

$$\oint_C (x+y)dx + (2x-z)dy + (y+z)dz$$

$$= \int_S (2\vec{i} + \vec{k}) \cdot \vec{k} \, dS \quad (\because \text{by Stokes's theorem})$$

$$= \iint_R dx \, dy = \text{Area of } \Delta^c$$

$$= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

$$= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

(10) Evaluate $\oint_C \sin z \, dz - \cos y \, dy + \sin y \, dz$ where C is the boundary of rectangle for $0 \leq x \leq \pi$; $0 \leq y \leq 1$; $z=3$

Volume integral (Gauss Divergence theorem)

Any integral which is to be evaluated in volume region is called volume integral

Theorem (1)

Statement: If $\vec{F}$ be a continuously differentiable a vector function define a over a closed surface 'S'

$$\oint_S \vec{F} \cdot \vec{n} ds = \iiint_V \text{div } \vec{F} dV$$

where 'V' is volume region defined by S

NOTE: It is a relation b/w closed surface integral to a volume integral

(Pb) using divergence theorem evaluate $\oint_S \vec{F} \cdot \vec{n} ds$ where $\vec{F} = 4x\vec{i} - 2y\vec{j} + z\vec{k}$ & 'S' is the surface bounded by the region $x^2 + y^2 = 4$, $z=0$; $z=3$

Sol Given that,

$$\vec{F} = 4x\vec{i} - 2y\vec{j} + z\vec{k}$$

$$\text{Div } \vec{F} = \nabla \cdot \vec{F} = 4 - 2 + 1 = 3$$

By divergence theorem,

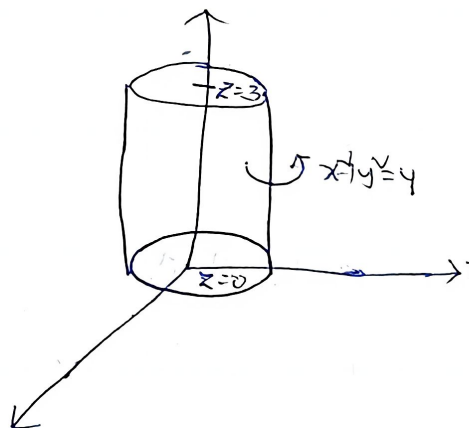
$$\oint_S \vec{F} \cdot \vec{n} ds = \iiint_V \text{div } \vec{F} dV$$

$$\therefore \oint_S \vec{F} \cdot \vec{n} ds = \iiint_V (4 - 2 + 1) dV$$

$$0 \leq z \leq 3$$

$$-\sqrt{4-y^2} \leq x \leq \sqrt{4-y^2}$$

$$-2 \leq y \leq 2$$



(Pb) evaluate above for

$$z=4$$

$$\begin{aligned} \vec{r} &= x\vec{i} + y\vec{j} + z\vec{k} \\ \frac{d\vec{r}}{dt} &= \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} + \frac{dz}{dt}\vec{k} \\ \vec{v} &= \frac{d\vec{r}}{dt} \\ \vec{v} &= \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} + \frac{dz}{dt}\vec{k} \end{aligned}$$

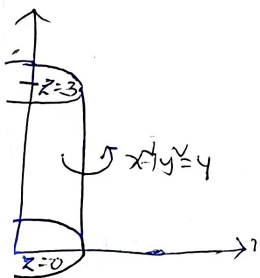
)

in volume region

integrate a vector

integral to a volume

finds where $\vec{F} = 4xz\vec{i} - 2yz\vec{j}$
region $x^2 + y^2 \leq 4$,



$$= \int_0^3 \int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} (4-4y+2z) dx dy dz$$

$$= 2 \int_0^3 \int_{-2}^2 (4-4y+2z) \sqrt{4-y^2} dy dz$$

$$= 2 \int_0^3 \left\{ \int_{-2}^2 (4+2z)(\sqrt{4-y^2}) dy + (-4) \int_{-2}^2 y \sqrt{4-y^2} dy \right\} dz$$

$$= 4 \int_0^3 (4+2z) \left(\int_0^2 \sqrt{4-y^2} dy \right) dz$$

$$= 4 \int_0^3 (4+2z) dz \times \left[\frac{y\sqrt{4-y^2}}{2} + \frac{z}{2} \sin^{-1}\left(\frac{y}{2}\right) \right]_0^2$$

$$= 4 \int_0^3 (4+2z) dz \times \pi$$

$$= 4\pi \left(4z + \frac{2z^2}{2} \right) \Big|_0^3$$

$$= 4\pi(12+9) = 84\pi$$

16. 2

evaluate $\iint \vec{F} \cdot \vec{n} ds$ over the entire surface of the region above the xy -plane bounded by the cone $z = \sqrt{x^2 + y^2}$ & the plane $z = 4$.

If $\vec{F} = 4xz\vec{i} + xy z\vec{j} + 3z\vec{k}$

$$\begin{aligned} \vec{F} &= 4xz\vec{i} + xy z\vec{j} + 3z\vec{k} \\ \vec{r} &= x\vec{i} + y\vec{j} + z\vec{k} \\ \vec{r}_x &= \vec{i} \\ \vec{r}_y &= \vec{j} \\ \vec{r}_z &= \vec{k} \\ \vec{n} &= \vec{r}_x \times \vec{r}_y + \vec{r}_y \times \vec{r}_z + \vec{r}_z \times \vec{r}_x \\ \vec{n} &= \vec{i} \times \vec{j} + \vec{j} \times \vec{k} + \vec{k} \times \vec{i} \\ \vec{n} &= \vec{k} + \vec{i} + \vec{j} \\ \vec{n} &= \vec{i} + \vec{j} + \vec{k} \end{aligned}$$

Verification method for (pb-1) question

Sol
$$\text{L.H.S} = \oint_S \vec{F} \cdot \vec{n} \, ds = \int_{S_1} \vec{F} \cdot \vec{n} \, ds + \int_{S_2} \vec{F} \cdot \vec{n} \, ds + \int_{S_3} \vec{F} \cdot \vec{n} \, ds$$

(i) For S_1 , $\vec{n} = -\hat{k}$; $ds = dx dy$

$$\therefore \int_{S_1} \vec{F} \cdot \vec{n} \, ds = \iint_{R_{xy}} (-z) \, dx dy = 0$$

(ii) For S_2 , $\vec{n} = \hat{k}$; $ds = dx dy$

$$\therefore \int_{S_2} \vec{F} \cdot \vec{n} \, ds = \iint_{R_{xy}} z \, dx dy = 9 \iint_{R_{xy}} dx dy = 9 \times \text{Area of circle}$$

$x^2 + y^2 = 4$

$= 9 \times 4\pi = 36\pi$

(iii) For S_3 : $x^2 + y^2 = 4$

here
$$\vec{n} = \frac{\nabla S_3}{|\nabla S_3|} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4x^2 + 4y^2}} = \frac{x\hat{i} + y\hat{j}}{2}$$

$$\therefore \int_{S_3} \vec{F} \cdot \vec{n} \, ds = \int_{S_3} \frac{4x^2 - 2y^3}{2} \, ds$$

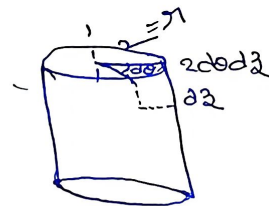
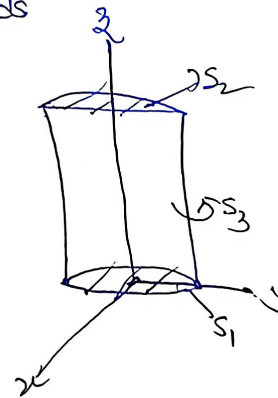
$$= \int_{S_3} (2x^2 - y^3) (2 \, d\theta \, dz)$$

$$= 2 \int_0^3 \int_0^{2\pi} (8\cos^2\theta - 8\sin^3\theta) \, d\theta \, dz$$

$$= 16 \int_0^3 \int_0^{2\pi} (\cos^2\theta - \sin^3\theta) \, d\theta \, dz$$

$= 48\pi$

$$\therefore \oint_S \vec{F} \cdot \vec{n} \, ds = 0 + 36\pi + 48\pi = 84\pi$$



(pb-2)

Sol Here

Di

By diver.

$$\oint_S \vec{F} \cdot \vec{n} \, ds$$

$$= \int_0^4$$

$$= 2 \int_0^4$$

$$= 2 \int_0^4$$

$$= 2 \times 2 \int_0^4$$

$$= 4 \int_0^4$$

$$=$$

17

Pb-2

$$\frac{x^2}{2} \sqrt{z^2 - y^2} + \frac{y^2}{2} \sin\left(\frac{y}{2}\right)$$

Sol Here $\vec{F} = 4xz\vec{i} + 2yz\vec{j} + 3z\vec{k}$

$$\text{Div } \vec{F} = 4z + xz + 3$$

By divergence theorem

$$\oint_S \vec{F} \cdot \vec{n} \, dS = \iiint_V \text{Div } \vec{F} \, dV$$

where V : $0 \leq z \leq 4$ (where V bounded by)

$$-\sqrt{z^2 - y^2} \leq x \leq \sqrt{z^2 - y^2}$$

$$-z \leq y \leq z$$

$$= \int_0^4 \int_{-z}^z \int_{-\sqrt{z^2 - y^2}}^{\sqrt{z^2 - y^2}} (4z + xz + 3) \, dx \, dy \, dz$$

$$= 2 \int_0^4 \int_{-z}^z (4z + 3) \sqrt{z^2 - y^2} \, dy \, dz$$

$$= 2 \int_0^4 (4z + 3) \int_{y=0}^z \sqrt{z^2 - y^2} \, dy \, dz$$

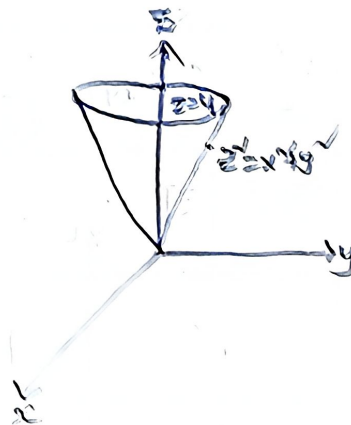
$$= 2 \times 2 \int_0^4 (4z + 3) \left[\frac{y\sqrt{z^2 - y^2}}{2} + \frac{z^2}{2} \sin\left(\frac{y}{z}\right) \right]_0^z \, dz$$

$$= 4 \int_0^4 (4z + 3) \left[\frac{(4z+3) z^2 \pi}{2 \cdot 2} \right] \, dz$$

$$= \pi \left[\frac{4z^4}{4} + 3 \cdot \frac{z^3}{3} \right]_0^4$$

$$= \pi [3^4 + z^3]_0^4$$

$$= 320\pi$$



This problem is unit-4

(pb) calculation of potential function from a irrotational field

Find the Potential Function ϕ $\nabla \times \vec{F} = \vec{0}$ & $\vec{F} = \nabla \phi$
 If $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$

Sol Given that

$$\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$$

$$\text{Curl } \vec{F}, (\nabla \times \vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2 - yz) & (y^2 - zx) & (z^2 - xy) \end{vmatrix}$$

$$= \vec{i} \left[\frac{\partial}{\partial y} (z^2 - xy) - \frac{\partial}{\partial z} (y^2 - zx) \right] - \vec{j} \left[\frac{\partial}{\partial x} (z^2 - xy) - \frac{\partial}{\partial z} (x^2 - yz) \right]$$

$$+ \vec{k} \left[\frac{\partial}{\partial x} (y^2 - zx) - \frac{\partial}{\partial y} (x^2 - yz) \right]$$

$$= \vec{i} [0 - 0] - \vec{j} [0 - 0] + \vec{k} (0 - 0)$$

$$= \vec{0} = \vec{F} \text{ is irrotational}$$

Then we can write $\vec{F} = \nabla \phi$

$$\Rightarrow (x^2 - yz) dx + (y^2 - zx) dy + (z^2 - xy) dz$$

$$= \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z}$$

$$\therefore \frac{\partial \phi}{\partial x} = x^2 - yz \quad \text{--- (i)} \quad \left| \quad \frac{\partial \phi}{\partial y} = y^2 - zx \quad \text{--- (ii)} \quad \left| \quad \frac{\partial \phi}{\partial z} = z^2 - xy \quad \text{--- (iii)} \right.$$

$$\text{From (i), } \phi = \frac{x^3}{3} - xyz + f(y, z)$$

$$\frac{\partial \phi}{\partial y} = 0 - xz + f'$$

$$\Rightarrow y^2 - zx = -xz + f' \quad (\because \text{from (ii)})$$

$$f' = y^2$$

$$f = \frac{y^3}{3} + g(z)$$

$$\therefore \phi = \frac{x^3}{3} + \frac{y^3}{3} +$$

$$\Rightarrow \frac{\partial \phi}{\partial z} = 0 + 0$$

$$\Rightarrow z^2 - xy = g'(z)$$

$$g'(z) = z^2$$

$$g(z) = \frac{z^3}{3}$$

$$\therefore \phi = \frac{x^3}{3} +$$

(or)

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy$$

$$= (x^2 - yz) dx$$

taking inter

$$\int d\phi = \int x^2$$

$$\phi = \frac{x^3}{3} + \frac{y^3}{3} +$$

$$\therefore \phi = \frac{x^3}{3} + \frac{y^3}{3} + g(z) - xy z$$

(iv)

$$\Rightarrow \frac{\partial \phi}{\partial z} = 0 + 0 + g'(z) - xy$$

$$\Rightarrow z^2 - xy = g'(z) - xy \quad (\because \text{from iii})$$

$$g'(z) = z^2$$

$$g(z) = \frac{z^3}{3} + C$$

sub in (iv)

$$\therefore \phi = \frac{x^3}{3} + \frac{y^3}{3} + \frac{z^3}{3} - xy z + C$$

(or)

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$= (x^2 - yz) dx + (y^2 - xz) dy + (z^2 - xy) dz$$

taking integration, we get

$$\int d\phi = \int x^2 dx + \int y^2 dy + \int z^2 dz - \left[\int xy dz + \int xz dy + \int yz dx \right] + C$$

$$\phi = \frac{x^3}{3} + \frac{y^3}{3} + \frac{z^3}{3} - xy z + C$$

$$\left(\downarrow \right. \\ \left. \text{lik } \int d(xy z) \right)$$

(p6) verify Gauss-divergence theorem for the vector field

(ii) for S_2

$\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ taken a rectangular

Parallelepiped $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$

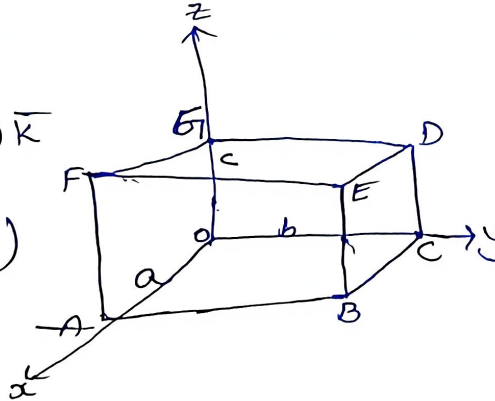
Sol Given that,

$$\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$$

$$\text{Div } \vec{F} = 2x + 2y + 2z = 2(x + y + z)$$

By divergence theorem,

$$\oint_S \vec{F} \cdot \vec{n} ds = \iiint_V \text{Div } \vec{F} dv$$



$$\Rightarrow \int_{S_1} + \int_{S_2} + \int_{S_3} + \int_{S_4} + \int_{S_5} + \int_{S_6}$$

(ABFE) (BCDE) DEFG OAEF OABC OEDG

$$= 2 \int_0^a \int_0^b \int_0^c (x + y + z) dz dy dx$$

(i) for S_1 , $\vec{n} = \vec{i}$, $ds = dy dz$

$$\therefore \int_{S_1} \vec{F} \cdot \vec{n} ds = - \int_0^b \int_0^c yz dy dz$$

$$= - \int_0^b \left[\frac{yz^2}{2} \right]_0^c dy$$

$$= - \int_0^b y \left(\frac{c^2}{2} \right) dy = - \frac{c^2 b^2}{4}$$

(18) For S_2 ; $\hat{n} = \vec{j}$; $ds = dx dz$